

Concept lattices for knowledge management

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The aim of the method presented in this paper is to support the acquisition of new knowledge and to enhance the interactions between knowledge workers. The approach chosen is to facilitate the sharing of those retrieval terms, which members of a community of practice have used to retrieve valuable information. The nature of information-seeking behaviour in on-line information sources is discussed and then the theory of formal concept analysis is introduced. It is subsequently shown how this theory can be applied to analyse the relations between documents and the retrieval terms that people use to access these documents.

The result is a concept lattice that contains information on a community's information-seeking behaviour. The concept lattice uncovers relational and contextual information. Retrieval phrases are put into relational context depending on how they are associated by the documents that are of interest to a community of users. The contention made here is that such 'usage-based' structures will provide natural and intuitive access to information sources for communities of users. It is shown how this approach can be used to facilitate the sharing of the retrieval vocabulary to support the acquisition of new knowledge and to enhance the interactions within a community of practice.

1. Introduction

There are now more than 200 million documents on World Wide Web (WWW) servers, along with millions more on corporate intranets. An increasing number of companies are realising that their intranets are valuable repositories of corporate information and that the WWW is a valuable public information resource. However, raw information in large quantities does not by itself solve business problems, produce value or provide competitive advantage. Knowledge management aims to exploit an organisation's information for greater productivity, new value, and increased competitiveness.

Due to a number of factors, including globalisation and the impact of the Internet, many organisations are increasingly geographically dispersed and organised around virtual teams. As noted in Goldman-Segall and Rao [1], such organisations need knowledge management and organisational memory tools that encourage users to understand each other's changing contextual knowledge and foster collaboration while capturing, representing and interpreting the knowledge resources of their organisations.

Enhanced corporate knowledge can be exploited in a variety of ways. By sharing and reusing current best practice, for instance, you can improve current business processes and eliminate duplication of effort. New business opportunities can be generated by collecting intelligence on markets and sales leads, and new products and services can be created, developed and brought to the market-place ahead of competitors.

The importance of knowledge sharing and reuse in knowledge management in order to share best practice and prevent duplication of effort has led to the emergence of the concept of communities of practice. Recent ethnographic studies of workplace practices indicate that the ways people actually work often differ fundamentally from the ways organisations describe that work in manuals, organisational charts and job descriptions [2]. The term 'community of practice' describes the formal groups where much knowledge sharing takes place and has been increasingly applied in the knowledge management context. Essentially a community of practice is a group of people who are '... peers in the execution of real work' (John Seely Brown, Xerox). They are typically not a formal team but an informal network, each sharing in part a common agenda and shared interests. In one example it was found that a lot of knowledge sharing among copier engineers took place through informal exchanges, often around the water cooler. As well as local, geographically based communities, trends towards flexible working and globalisation has led to interest in supporting global communities using Internet technology. The challenge for organisations is to support such communities and make them effective.

2. Relating information to foster and share knowledge

Building an intranet-based store of information is not sufficient for knowledge management — the relationships within the stored information are vital. These relationships cover such diverse issues as relative importance, context,

sequence, significance, causality and association. Traditional search engines return ranked retrieval lists that do not offer any information on the semantic relationships among documents. Knowledge workers have to spend a substantial amount of their time browsing and reading to find out how documents are related to one another and where each falls into the overall structure of the problem domain. Yet only when knowledge workers begin to locate the similarities and differences among pieces of information do they move into the essential part of their work — building relationships to create new knowledge.

Information retrieval traditionally focuses on the relationship between a given query (or user profile) and the information store. On the other hand, exploitation of interrelationships between selected pieces of information can put otherwise isolated information into a meaningful context. The implicit structures revealed help users make use of and manage information more efficiently [3]. Tools are needed that integrate the resources dispersed across Web pages into a coherent corpus of interrelated information. Previous research (see for example Hearst [4]) has largely focused on integrating heterogeneous databases and knowledge bases, which represent information in a highly structured way, often by means of formal languages. In contrast, the World Wide Web consists to a large extent of loosely structured natural language texts.

Ontologies can open alternative ways to cope with heterogeneous representations of Web resources. An ontology, in the sense used here, can be thought of as a 'concept map' or, more formally, the specification of a conceptualisation of a domain. Provided with an ontology meeting the needs of a particular community of practice, knowledge management tools can arrange knowledge assets into the predefined conceptual classes of the ontology. If the conceptual classes are defined such that they exhibit the resemblance and differences essential to knowledge workers, it is possible for them to draw conclusions about relationships and implications.

Ontologies are held to facilitate communications between knowledge workers within a community, since meaningful interaction can only occur when people share a common interpretation of the vocabulary used in their communication. This approach to knowledge sharing is based on the assumption that if a domain model is explicitly specified and agreed upon by a number of agents, it is easier for them to share and reuse knowledge.

However, different communities of practice need to access information via a view that is appropriate to them. Engineers and sales people, for example, may have a very different perspective on how different pieces of information relate to one another. The challenge is therefore to present a natural and intuitive interface on to an information repository for each class of user, rather than imposing a single conceptualisation or 'world view' on everybody.

Many current applications of ontologies for knowledge management are implicitly committed to the view that knowledge workers should all agree on one meaning for each word and use it only with that meaning [5]. However, we simply cannot make people agree in this way, even if we could set up a supervising scheme to dragoon people into using each word with its predefined meaning [6]. The meaning will depend on the listener, on the speaker, on their entire experience of the language, on their knowledge of one another, and on the whole situation. People reading the same text may interpret it differently, and hence act on it differently (see Meadow [6]). It is the humans who give the symbols meaning and significance.

The full meaning of a word does not appear until it is placed in its context. Meanings of words are always shifting and changing. As Hayakawa puts it: 'No word ever has exactly the same meaning twice' [7]. In practice, community languages emerge through acquisition of new information and interactions among the members.

If the meaning of a word depends on use, interpreter, and context then the most promising way to capture the meaning of a community's vocabulary is to analyse how members use that vocabulary to accomplish their goals, and to interact with other members of the community. Since it is impossible to track users in all their communicative activities, the focus here will be on people's information-seeking behaviour so as to capture and analyse the vocabulary they use during this process.

The hypothesis is that the change of meanings or the emergence of new concepts will affect the way people use knowledge repositories to solve their problems. This hypothesis is based on the use theory of meaning, according to which the meaning of a concept is its use in communication and more generally in social interactions.

The next section discusses the nature of information-seeking behaviour in on-line information sources and then looks at formal concept analysis and how this technique can be exploited to derive a shared concept lattice from users' information-seeking behaviour. It is our contention that such a 'usage-based' concept lattice will provide the structural basis for natural and intuitive access to information sources for communities of users.

3. Information-seeking behaviour

Information seeking is a communication process. Authors or creators of information communicate with readers, but indirectly and with a possibly long time lag between creation of a text and its delivery to the reader [6]. Documents are the primary means of information exchange in this process, in that they convey and communicate knowledge. Documents contain concepts, and information on the relationships between concepts.

People actively seek information by formulating and adjusting queries. They spend large amounts of time trying to express their information needs in ways that eventually retrieve the relevant documents. During this process they have to make judgements about the usefulness of texts by engaging with them. They interpret the text and construct meaning. They assign meanings to words to reformulate their queries using terms that represent their information needs in the technical language used in the documents containing the desired information.

There are various systems on the Internet that seek to exploit information-seeking behaviour. For example, Direct Hit [8] is a system which analyses the retrieval behaviour of information seekers to determine the relevance a page's content has to a searcher's query. It compiles its retrieval lists by tracking the way search engines are used. Direct Hit captures users' retrieval behaviour by tracking keyword searches and measuring what pages users select from search results and roughly how long they spend visiting these pages. Subsequent searches by users will then automatically list the top ten most-frequented sites checked out by others looking for the same keyword. The system also displays narrower variants of the original search query, and displays the ten most popular alternatives. Users can then refine their searches based on the paths others have followed.

This retrieval behaviour based approach is reminiscent of collaborative filtering, but there are significant differences. Systems such as Ringo [9] and ProRate [10] make recommendations based on explicit user responses. Retrieval behaviour based filtering, on the other hand, passively aggregates implicit user relevance judgements around particular search topics. Collaborative filtering is limited by the relatively low number of subjective responses, whereas retrieval behaviour based filtering has the advantage of tapping into a voluminous database of searchers' retrieval behaviours.

A retrieval dialogue can be seen as a negotiation between information seeker and provider to agree on a shared vocabulary. Successful retrieval requires successful negotiation that yields an implicit agreement between the information seeker and the provider on a shared vocabulary that provides effective retrieval terms. Users can make use of these concepts to retrieve new information and to communicate newly obtained knowledge to members of their community. The acquisition of new information and the interactions among the community members means that community languages constantly evolve.

The aim of the approach presented in this paper is to facilitate the sharing of people's retrieval vocabulary to support the acquisition of new knowledge and to enhance the interactions among community members. This goal cannot be achieved by only recording people's retrieval vocabulary, as Direct Hit does. It is crucial also to capture

the relations between retrieval terms and to characterise the context in which they were used.

In the research reported here, the theory of formal concept analysis is used to analyse the relations between documents and the retrieval terms that people use to retrieve these documents. The underlying assumption is that retrieval phrases are linked by the documents they jointly retrieve. The relations obtained in this process can be seen as grounded in the documents accepted by a community.

4. Formal concept analysis

Derived from Birkhoff's work in lattice theory, formal concept analysis was introduced by Rudolf Wille [11] in 1982. The subsequent descriptions of formal concept analysis mainly follow Davey [12], Kent [13], Priss [14] and Ganter [15], which should be consulted for more

Formal concept analysis is based on the philosophical understanding of a concept as a unit of thought consisting of two parts — the extent and the intent. The extent consists of all objects belonging to the concept while the intent is the collection of all attributes shared by the objects.

Formal concept analysis works within a specific formal context in which the objects and attributes are fixed. A formal context is defined as a triple (G, M, I) consisting of a set of formal objects G , a set of formal attributes M , and a relation $I \subseteq G \times M$ between G and M . A given formal context can be represented by a cross table which has a row for each formal object g , a column for each formal attribute m and a cross in the row of g and the column of m if $(g, m) \in I$.

Table 1 shows an example formal context. Although somewhat simplistic, it is quite useful for illustrative purposes. The formal objects are girl, woman, boy, and man while the formal attributes are female, juvenile, adult and male. A formal concept of this context will consist of an ordered pair (A, B) , where A is a subset of the four formal objects and B is a subset of the four formal attributes. The intent should contain precisely those attributes shared by all objects in the extent, and conversely the extent should contain precisely those objects sharing all attributes in the intent. Both its extent and its intent therefore determine a formal concept.

Table 1 Cross table of example formal context.

	female	juvenile	adult	male
girl	x	x		
woman	x		x	
boy		x		x
man			x	x

In order to formalise the definition of a formal concept we introduce the derivation operator. In a formal context

(G, M, I) the set of all common attributes of a set $A \subseteq G$ of objects is denoted by:

$$A' := \{m \in M \mid (g, m) \in I \text{ for all } g \in A\}$$

Analogously the set of all common objects of a set $B \subseteq M$ of attributes is:

$$B' := \{g \in G \mid (g, m) \in I \text{ for all } m \in B\}$$

For instance, in the example formal context the following derivations hold: $\{\text{girl}\}' = \{\text{female}, \text{juvenile}\}$ $\{\text{male}, \text{adult}\}' = \{\text{man}\}$. Using the derivation operator, a formal concept of the context (G, M, I) can be defined as a pair (A, B) where $A \subseteq G$, $B \subseteq M$, $A' = B$ and $B' = A$, A is the extent and B is the intent of the concept (A, B) .

In the example context $(\{\text{woman}, \text{man}\}, \{\text{adult}\})$ is a formal concept because $\{\text{woman}, \text{man}\}' = \{\text{adult}\}$ and $\{\text{adult}\}' = \{\text{woman}, \text{man}\}$ hold. Note that all possible pairs are formal concepts, since (A, B) must satisfy the constraints $A' = B$ and $B' = A$, which in practice prove to be very restrictive constraints. The set of all formal concepts of the formal context (G, M, I) is denoted by $B(G, M, I)$.

The formal concepts of a given formal context can be ordered by the generalisation-specialisation relation. A formal concept (A_1, B_1) is regarded as being less general than a formal concept (A_2, B_2) if the extent A_1 of (A_1, B_1) is contained in the extent A_2 of (A_2, B_2) . The formal concept (A_1, B_1) is called subconcept of (A_2, B_2) and (A_2, B_2) is called superconcept of (A_1, B_1) .

The subconcept-superconcept relation defines an order on the set of formal concepts by:

$$(A_1, B_1) \leq (A_2, B_2) \Leftrightarrow A_1 \subseteq A_2$$

For example $(\{\text{woman}\}, \{\text{adult}, \text{female}\})$ is a subconcept of $(\{\text{woman}, \text{man}\}, \{\text{adult}\})$ because the extent $\{\text{woman}\}$ is contained in the extent $\{\text{woman}, \text{man}\}$ or expressed formally $(\{\text{woman}\}, \{\text{adult}, \text{female}\}) \leq (\{\text{woman}, \text{man}\}, \{\text{adult}\})$.

The resulting ordered set of all formal concepts forms a complete lattice, which is known as concept lattice. Graphically, a concept lattice can be visualised by a line diagram with nodes representing formal concepts and edges representing the subconcept-superconcept relations between formal concepts. It is not necessary to write the full extent and intent for each formal concept. A much simpler, reduced labelling is achieved if each object and each attribute is entered only once in the diagram.

With reduced labelling an object label is attached below the node of the smallest concept whose extent contains the object, while an attribute label is attached above the node of the largest concept whose intent contains the attribute.

A concept lattice with reduced labelling is interpreted by using the following simple rules:

- the extent of a concept node is the set of all objects that label this node or other nodes on a path going downward,
- the intent of a concept node is the set of all attributes that label this node or other nodes on a path going upward,
- all concept nodes above a node labelled by an object have the object in their extent,
- all concept nodes below a node labelled by an attribute have the attribute in their intent.

The concept lattice generated from the example context is shown in Fig 1.

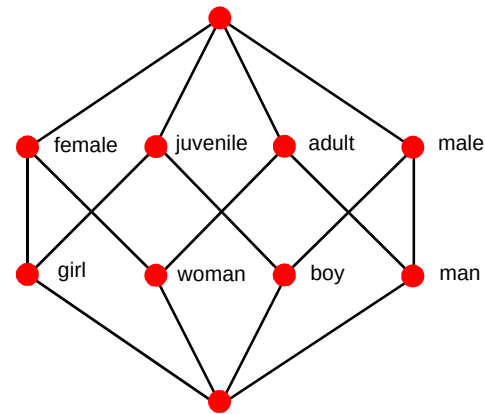


Fig 1 Concept lattice generated from example formal context.

Both objects and attributes generate formal concepts. The formal concept generated by an object $g \in G$ is defined by $\gamma(g) := (\{g\}', \{g\})$. In the subconcept-superconcept order, this is the smallest formal concept having g in its extent. Dually, the formal concept generated by an attribute $m \in M$ is defined by $\mu(m) := (\{m\}', \{m\})$. This is the largest formal concept having m in its intent. The formal concepts $\gamma(g)$ and $\mu(m)$ are called object concept of g and attribute concept of m , respectively.

5. Concept analysis of retrieval vocabularies

5.1 Intelligent searching

This section describes how formal concept analysis is employed to analyse a community's retrieval vocabulary. The first case considered here is where a member of a community of practice is searching for information and wishes to exploit the community's previous search experience and retrieval vocabulary. The retrieval vocabulary is captured by tracking the search results links that users click on and by measuring how long users stay at each site. Additionally, the word phrases that people use for formulating their queries are recorded.

A document is represented as a list of terms. The terms are the word phrases that occur within the queries of those users who actually went to investigate the document. Each term is assigned weights in the range 0 to 1. A term's weight for a document expresses the probability that users who retrieved the document by using the term in their queries, found the document to be relevant. In other words a term's weight for a document represents the probability that the term is a hit phrase of the document, with a hit phrase being a word phrase whose use within a query leads users to a relevant document. A value of 1 denotes complete certainty that the term is a hit phrase and 0 denotes complete certainty that the term is not a hit phrase, with intermediate values representing corresponding degrees of belief.

A term's weight for a document is calculated by determining the percentage of users that, upon using the term in their queries, investigated the content of the document. The weight is further adjusted according to the amount of time that those users spent looking at the document. A term's weight can also be seen as the degree to which the term represents the meaning of the document content.

Formally, these data can be represented as a matrix of documents by terms whose m rows and n columns correspond to m documents and n terms, respectively. The elements of the document-term matrix are the weights of each term for a particular document, that is:

$$Y = \begin{bmatrix} y_{11} & \cdots & y_{1n} \\ \vdots & & \vdots \\ y_{m1} & \cdots & y_{mn} \end{bmatrix}$$

where y_{ij} denotes the weight assigned to term T_j for document D_i .

Assume the user has submitted a query and a search engine has returned a list of documents matching the query. The formal concept analysis of the data starts with the

creation of a formal context. The formal objects of the formal context are the documents D_i that were returned as the top ten in the retrieval list. The set of these documents is denoted by D . Using the retrieval behaviour database the algorithm determines the members of the community that investigated the documents in D . The hit phrases T_j that those members used to retrieve the documents in D , constitute the formal attributes of the formal context. The set containing these attributes is denoted by T .

I is the binary incidence relationship between D and T . $(D_i, T_j) \in I$ asserts that term T_j is a hit phrase of document D_i . The relationship I is derived from the matrix Y by comparing the values of Y with a threshold t : $I := \{(D_i, T_j) \in D \times T \mid y_{ij} > t\}$, where t is a predefined threshold. The value t is somewhat arbitrarily fixed in the current system at the constant value of 0.2.

The cross table of the resulting formal context has a row for each document in D , a column for each term in T and a cross in the row of D_i and the column of T_j if the corresponding weight y_{ij} exceeds the value of the threshold t . To include the user's currently submitted query an additional column is added with a cross for each document in D . After the formal context is constructed, formal concept analysis is applied to produce the concept lattice.

To consider an example, assume that a user enters the query 'data and information management'. After processing the query the retrieval engine returns a list of possibly relevant documents. The titles of the top ten documents returned are shown next:

- A: Information agents for the WWW
- B: Jasper: communicating information agents
- C: Data mining for marketing
- D: ALADIN — Applied Logic For Advanced Data Mining
- E: Communities of interest
- F: Data management research
- G: Autonomous management of distributed information systems
- H: Data mining research
- I: Networked information management
- J: ProSearch: a profile-based search engine

According to the tracked retrieval data, these documents were investigated by users who retrieved these documents

using queries that contained the following phrases and words:

Phrase/word	Abbreviation
Data management	(DM)
Information management	(IM)
Knowledge management	(KM)
Distributed systems	
Data mining	
Autonomous systems	
Logic	
Marketing	

The document-term matrix with the weights assigned to the terms for each document is given in Table 2.

The cross table (Table 3) representing the formal context is obtained by applying the certainty threshold of 0.2. A cross in the row of D_i and the column of T_j indicates that T_j is believed to be a hit phrase of D_i . Finally, the column for the user's currently submitted query 'data and information management' is added. The query is abbreviated as DIM.

The formal concepts of the example contexts are shown as extent-intent pairs, where the extent is a subset of the top ten documents and the intent is a subset of the terms.

({A, B, C, D, E, F, G, H, I, J}, {DIM})
 ({I, F, E, B, J, A, G}, {DIM, IM})
 ({F, H, B, J, E, C, D}, {DIM, DM})
 ({F, B, J, E}, {DIM, DM, IM, KM})
 ({I, A, G, B, J}, {DIM, IM, Distributed S})
 ({B, J}, {DIM, IM, KM, Distributed S})
 ({A, G}, {DIM, IM, Distributed S, Autonomous S})
 ({H, E, C, D}, {DIM, DM, Data Mining})
 ({C}, {DIM, DM, Data Mining, Marketing})
 ({D}, {DIM, DM, Data Mining, Logic})
 ({E}, {DIM, DM, IM, KM, Data Mining})

The intent of each formal concept contains precisely those hit phrases shared by all documents in the extent. Conversely, the extent contains precisely those documents sharing all hit phrases in the intent.

Information on the context of a term can be obtained by analysing the formal concept that the term generates. The term 'autonomous systems' for example generates the following formal concept:

Table 2 Document-term matrix.

	DM	IM	KM	Distributed systems	Data mining	Autonomous systems	Logic	Marketing
A	0.0	0.6	0.1	0.6	0.0	0.6	0.0	0.0
B	0.3	0.4	0.8	0.3	0.0	0.1	0.0	0.0
C	0.3	0.1	0.1	0.0	0.9	0.0	0.2	0.8
D	0.3	0.0	0.0	0.0	0.8	0.0	0.9	0.1
E	0.3	0.4	0.8	0.0	0.4	0.0	0.0	0.0
F	0.8	0.7	0.3	0.2	0.0	0.1	0.0	0.0
G	0.0	0.3	0.0	0.6	0.0	0.8	0.0	0.0
H	0.3	0.1	0.0	0.2	0.7	0.0	0.2	0.2
I	0.2	0.5	0.0	0.5	0.0	0.1	0.0	0.0
J	0.3	0.6	0.7	0.4	0.0	0.0	0.0	0.0

Table 3 Formal context.

	DIM	DM	IM	KM	Distributed systems	Data mining	Autonomous systems	Logic	Marketing
A	X		X		X		X		
B	X	X	X	X	X				
C	X	X				X			X
D	X	X				X		X	
E	X	X	X	X		X			
F	X	X	X	X					
G	X		X		X		X		
H	X	X				X			
I	X		X		X				
J	X	X	X	X	X				

({A, G}, {DIM, IM, Distributed Systems, Autonomous Systems})

The generated concept specifies that the documents A and G retrieved by the term 'autonomous systems' were also retrieved by all the other terms in the intent. The terms in the intent are therefore linked to the term 'autonomous systems' in that they were all used to retrieve the documents in the extent.

The concept lattice, showing the partially ordered set of concepts is shown in Fig 2. The top formal concept represents the user's currently submitted query.

The line diagram of the lattice has the minimal set of edges necessary; all other edges can be derived by using reflexivity and transitivity. Documents and terms label the node that represents the formal concept they generate. A document label is attached below the node of the smallest concept whose extent contains the document, while a term label is attached above the node of the largest concept whose intent contains the term. The extent and the intent of concept can be read off the line diagram by applying a few simple rules. The extent of a concept node is the set of all documents that label this node or other nodes on a path going downward. The intent of a concept node is the set of all hit phrases that label this node or other nodes on a path going upward.

All concept nodes above a node labelled by a document have the document in their extent. All concept nodes below a node labelled by a hit phrase have the hit phrase in their intent. The extent of the concept node labelled by the term 'knowledge management' for example is easily found by first collecting the document F labelling this concept node and then collecting the documents B, J, E labelling the concepts on a path going downward.

The intent of this concept is found by first collecting the term 'knowledge management' and by going upward to collect the terms 'data management', 'information management', and 'data and information management' labelling the three concepts found on paths going upward. The resulting extent-intent pair of this concept is ({F, B, J, E}, {DIM, IM, KM}).

The concept generated by the term 'data mining' is a subconcept of the concept generated by the term 'data management', for the extent of the former concept is contained in the extent of the latter concept. All documents retrieved by the hit phrase 'data mining' were also retrieved by the hit phrase 'data management', suggesting that within the given formal context 'data mining' is a more specific term than 'data management'.

In a similar way the concept lattice seems to suggest that the normally very general terms 'marketing' and 'logic' are more specific terms than 'data mining'. This contradicts our experience. To resolve this contradiction it has to be noted that formal concept analysis works within a specific context in which both the set of objects and the set of attributes are fixed. The concept lattice always refers to that meaning of a term which it assumes within the given context. In the example context, all documents retrieved by the hit phrases 'logic' and 'marketing' were also retrieved by the hit phrase 'data mining', i.e. all documents with the hit phrases 'logic' and 'marketing' also have the hit phrase 'data mining'. This indicates that in the given context the terms 'logic' and 'marketing' are not to be taken in the sense of denoting the respective subjects but rather to be interpreted as representing the application of logic and marketing within data mining. This example demonstrates that due to the variability of meanings, the occurrence of any single word does not necessarily mean that the implied subject is described.

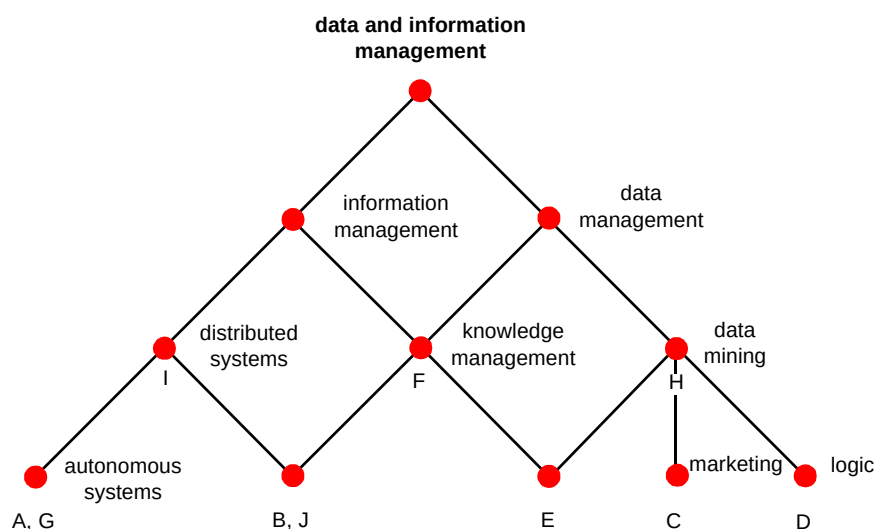


Fig 2 Concept lattice of retrieval results.

Additional information on the contents of the documents retrieved by the hit phrase 'data mining' is given by showing that document D is related to logic, document C related to marketing and document E related to knowledge management.

By navigating the concept lattice the user can exploit the community's previous information-seeking experience. A newly retrieved document is described by phrases that were used by the community to retrieve apparently relevant information. The presented retrieval phrases also reveal the community's interest related to the user's current information need.

The concept lattice uncovers relational and contextual information. Retrieval phrases are put into relational context depending on how they are associated by the documents that are of interest to the community. Formerly unknown relations can be revealed. The phrases 'data mining' and 'logic' for example are shown as related not simply because people entered queries containing both terms, but rather because these terms share a similar retrieval context. If people use these terms in conjunction to formulate queries then they are already aware of relations between those terms. The reason for constructing the concept lattice is to make people aware of hidden relationships that emerge from the community's retrieval behaviour.

5.2 Building lattices for knowledge repositories

The retrieval vocabulary analysis presented above is only applicable when documents were already retrieved and investigated by information seekers. In order to alleviate this restriction, a phrase extraction method was developed. This method tries to extract phrases that enable a user presented with the extracted phrases, to draw conclusions about the content of the document [16]. The use of phrase extraction will allow the method described above to be used on any set of documents submitted to the system, rather than being limited to queries made by particular users.

Phrase extraction is achieved by a novel method (patent application pending) that identifies significant phrases and sub-phrases within a block of text. The method uses a series of heuristic steps, tuned to give best overall performance, together with part-of-speech information.

Preference is given to longer phrases, but the maximum acceptable phrase length may be specified. The essence of the method is to propagate the significance of shorter phrases into phrases of greater length. The resulting phrases can often give a flavour of a document without occurring particularly frequently within it.

For example, if a document contains several instances of the words 'bread' and 'butter', and also includes the phrase 'bread and butter', then 'bread and butter' is likely to be

identified as a key phrase, even though it may occur infrequently.

The processing steps used in this method appear to give a better subjective performance than is possible with straightforward statistical analysis techniques. Although the resultant phrases can feed into further processing stages, they are particularly suitable for presentation purposes. They can also complement automatically produced document abstracts.

6. Conclusions

Current applications of ontologies for knowledge management are implicitly committed to the view that knowledge workers should all agree on one meaning for each word [17]. However, the meaning of a word depends on use, interpreter, and context. It was therefore suggested that the meaning of a community's vocabulary be captured by analysing how members use that vocabulary to search on-line information sources.

The theory of formal concept analysis was employed to derive a concept lattice from the usage a group of users make of information resources. Seeing how documents are related to one another, the user can locate the similarities and differences between pieces of information. Presentation of interrelationships between selected pieces of information puts otherwise isolated information into a meaningful context. The implicit structures revealed help users to access information more efficiently. By navigating the concept lattice, the user can exploit the community's previous information-seeking experience. This approach has the potential to support the emergence of new knowledge by identifying concept relations, making these explicit and enabling users to inspect these concept relations.

Concept lattices are not intended to build or substitute traditional static ontologies, rather they aim to support specifications of less rigorous relations, or associations, which might be more intuitive to knowledge workers and lead to more interesting links via associations.

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